



Optimal representation of spring phenology on photosynthetic productivity across the northern hemisphere forests

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ABSTRACT

Accurate simulation of the onset of spring is crucial for predicting the photosynthetic productivity of forest ecosystems. Nonetheless, the potential of phenology simulations on predictions of forest gross primary productivity (GPP) remains poorly understood, with previous studies generally focused on predicting phenology dates themselves or on limited scales. Here, we constructed a framework through the critical process of leaf growth to couple a terrestrial biosphere model—FORCCHN2 with five widely used phenology models, including one-phase (involving heat forcing), two-phase (considering chilling demand), and multi-phase models (integrating photoperiod effects) through the critical process of leaf growth. We evaluated the GPP performance from the multiple forest biomes in 74 eddy covariance (EC) sites and compared the total trends to satellite observations across the northern hemisphere forests. The predictions of the five models could reproduce 62.41–67.24 % variations of the GPP observations in all EC sites. In the deciduous broadleaf forest sites, we found the two-phase and multi-phase coupled models performed better than the one-phase coupled model. Chilling may play an essential role in controlling photosynthetic activity in deciduous broadleaf forests. Moreover, the multi-phase coupled model integrated photoperiod effects performed best at capturing GPP changes of the whole northern hemisphere forests. The predictions also showed the GPP in the northern hemisphere ranged from 21.68 to 26.06 Pg C year⁻¹, and the total GPP showed an increasing trend. This study provides local and regional available evidence for the impact of phenology on photosynthesis with climate warming. The results highlight the necessity for enhancing understanding of the phenology factors on GPP and integrating the appropriate phenology representation to improve predictions of photosynthetic productivity for forests, especially in prediction to large scales and long-term trends.

1. Introduction

Forest trees fix CO₂ as organics through photosynthesis, and this carbon flux is also known as the gross primary productivity (GPP) at the ecosystem level (Beer et al., 2010; Anav et al., 2015). Climate warming induced changes in spring phenology are considered to be a primary reason for the increase in terrestrial vegetation GPP (Piao et al., 2017). Previous studies have developed multiple spring phenology models, mainly focusing on the simulation of phenology dates, to express the response and feedback of vegetation to climate change (Cannell and

Smith, 1983; Chuine, 2000; Caffarra et al., 2011). These models are used to investigate plant phenology responses to future climate change and integrate with terrestrial biosphere models (TBMs) to explore carbon dynamics from individual to global scales. However, some vegetation phenology models are considered to be overly simplistic and produce considerable uncertainties when coupled with TBMs, leading to large biases in estimating GPP (Ryu et al., 2008; Randerson et al., 2009; Richardson et al., 2012). Clearly, to promote large-scale GPP modeling by terrestrial biosphere models and to increase our understanding of the global carbon cycle in the context of climate change, an investigation of

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